

5 Ans: C

Kinetic Energy = $p^2 / 2m$

Since both M and m have the same Kinetic Energy, $p^2 \propto m$

$$\therefore \frac{p_M^2}{p_m^2} = \frac{M}{m}$$

$$\frac{p_M}{p_m} = \sqrt{\frac{M}{m}}$$

6 Ans: C